

The inverse self-commutator cost at inertia (m, n) : ordered-layering bounds, interlacing chains, and the Horn tiles that certify them

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Abstract

For a traceless Hermitian $F \in M_d(\mathbb{C})$ let $\kappa_d(F) = \frac{1}{2} \min\{\|C\|_{HS}^2 : CC^* - C^*C = 2F\}$. On the inertia- (m, n) stratum (m positive and n negative eigenvalues, z zeros) we prove three results valid for all m, n, z : a dimension-free lower bound $\kappa_d \geq \text{cost}(L)$ for every ordered full-capacity layering L , certified by Lidskii–Wielandt and Weyl inequalities alone; an exact additive formula on the aligned-spike cone, where an optimizer is a direct sum of weighted shifts; and the closed form $\kappa_d = \text{cost}(L_k)$ on explicit full-dimensional regions R_k , $1 \leq k \leq m - n + 1$, obtained from one rank-one interlacing step. Open question 1 of the inertia- $(m, 2)$ paper is refuted by an exact weak-duality certificate: at inertia $(5, 4)$ a Horn tile with $n = 4$ boxes is necessary, and one exposed form needs a two-row shape. The $d = 8$ rank-onset example is explained by a late negative. Conjectures for general (m, n) are stated with their numerical evidence.

1. Introduction

The quantity $\kappa_d(F)$ is the cheapest realisation, in Hilbert–Schmidt norm, of a traceless Hermitian F as a self-commutator $\frac{1}{2}(CC^* - C^*C)$. Its theory has been developed in a line of records of the AI Research Record archive, cited by descriptive keys (identifiers in the reference list); everything used from them is restated here. [FourLevel] reduced κ_d to a finite linear program over Horn inequalities (the *Horn program* (2.1)); [OneSpike] proved the one-spike formula $\kappa_d = \sum_j j a_j$ when one sign has multiplicity one; [RankOnset] proved that the minimal optimal rank r_* equals the inertia bound $r_0 = \max(n_+, n_-)$ for $d \leq 7$ and exhibited the first spectrum, $d = 8$ of inertia $(4, 4)$, with $r_* > r_0$. The published inertia- $(m, 2)$ paper [Mtwo] gave a dimension-free closed formula for two negative eigenvalues (Lidskii–Wielandt, Weyl and one Pieri-type Horn triple below, 2×2 -block weighted shifts above) and asked in its Open question 1, quoted verbatim in Section 6, what survives at inertia (m, n) . The unpublished inertia- $(m, 3)$ manuscript [Mthree] answered for three negatives and $m \leq 9$ by computer-assisted “tiling” certificates and 3×3 -block shifts, proved a dimension-free lower bound for its ordered family A, and found a padding phase.

Contribution. We treat general inertia (m, n) . Theorem 1 proves, for every $m, n \geq 1$ and $z \geq 0$, that the cost of every *ordered full-capacity layering* — positives in decreasing order, every non-final layer holding exactly as many positives as there are negatives before it, any order of the negatives — is a lower bound for κ_d , certified by Lidskii–Wielandt and Weyl inequalities alone. It contains Theo-

rem 2 of [Mthree] and the F_k bounds of [Mtwo] and extends them to schedules with inversions of the negatives; a rearrangement lemma shows that the best ordered bound is attained by a nondecreasing schedule, and an exact counterexample shows that full capacity cannot be relaxed. Theorem 2A characterises the layerings whose block-shift chain can be solved *diagonally*: exactly those carrying a balanced path system; the cost is then additive, $\kappa_d = \sum_i \sum_j j a_j^{(i)}$, and an optimizer is a direct sum of one-spike weighted shifts, while alignment is a genuine restriction ($23 < 24$ in an explicit example). Theorem 2B gives, for every $n \geq 2$ and $1 \leq k \leq m - n + 1$, a full-dimensional region R_k on which κ_d equals the explicit form $\text{cost}(L_k)$, by a chain that is diagonal except for one rank-one step whose feasibility is the converse of Cauchy interlacing [Golub [Go73], Chu [Chu98]]; for $n = 2$ the regions are the F_k chambers of [Mtwo]. Theorem 3 refutes the first half of Open question 1 of [Mtwo]: at inertia $(5, 4)$ an exposed form is certified by no combination of Horn inequalities with at most $n - 1 = 3$ boxes (an exact rational weak-duality certificate) but by a tiling with a 4-box Pieri tile. The second half (blocks of size n attain) is supported numerically on every stratum tested and stated as a conjecture. Section 7 explains the $d = 8$ example of [RankOnset]: its two optimal forms are costs of feasible block shifts in which the largest negative enters a late layer, which forces rank 5 without padding. Section 8 collects the numerical structure of the exposed forms on 14 strata (up to $d = 12$) and states a conjecture; Section 9 lists open questions.

Status, stated honestly. Theorems 1, 2A, 2B and 3 are proved (Theorem 3 with an exact rational computation); Section 8 is numerical and Section 8.3 conjectural, and every numerical statement is labelled as such. An independent adversarial review re-derived every proof and re-implemented every computational ingredient (Section 10); its corrections — the range of k in Theorem 2B, the Weyl tiles of the box-4 certificate, a counterexample to a conjectured “Pieri shapes suffice” statement — are incorporated. Literature searches (Section 10) found no prior closed formula on an inertia stratum and no prior use of the interlacing converse inside a block-shift chain; the nearest works [JOS13, AS15] bound different objectives over factorisations $A = [B, C]$. A null search does not certify priority.

2. Setting

Throughout $\|X\|^2 = \text{Tr } X^*X$ and $F \in M_d(\mathbb{C})$ is Hermitian and traceless with ordered spectrum $\lambda_1 \geq \dots \geq \lambda_d$. On the inertia- (m, n) stratum

$$\lambda = (a_1 \geq \dots \geq a_m > 0, 0^z, -b_n, \dots, -b_1), \quad b_1 \geq \dots \geq b_n > 0, \quad \sum_j a_j = \sum_i b_i =: P, \quad d = m+n+z.$$

The spectrum index of a_j is j , that of $-b_i$ is $d + 1 - i$, and the zeros occupy $m + 1, \dots, m + z$. We write $A_u := \sum_{j \geq u} a_j$ ($A_{m+1} = 0$) and $B' := b_2 + \dots + b_n$. By $F \mapsto -F$ (which replaces C by C^*) every statement transfers to inertia (n, m) .

Given C with $CC^* - C^*C = 2F$, put $R = CC^*/2$, $S = C^*C/2$: they are positive semidefinite with the same spectrum $s_1 \geq \dots \geq s_d \geq 0$, $R - S = F$, $\frac{1}{2}\|C\|^2 = \sum_t s_t$, and conversely any such pair comes from $C = U\sqrt{2S}V^*$. Hence ([FourLevel] Theorem 2.2, [RankOnset] Proposition 2.2, [Mtwo] (2.1))

$$\kappa_d(F) = \min \left\{ \sum_{t < d} s_t : s_1 \geq \dots \geq s_{d-1} \geq s_d = 0, (s, -s^{\text{rev}}, \lambda) \text{ Horn-feasible} \right\}, \quad (2.1)$$

where Horn-feasible means that $\alpha = (s_1, \dots, s_{d-1}, 0)$, $\beta = (0, -s_{d-1}, \dots, -s_1)$ and $\gamma = \lambda$ satisfy every Horn inequality $\sum_{k \in K} \gamma_k \leq \sum_{i \in I} \alpha_i + \sum_{j \in J} \beta_j$, $(I, J, K) \in T_r^d$ [Ho62, Kl98, KT99, Fu00]. The normalisation

$s_d = 0$ is legitimate because subtracting $s_d \mathbf{1}$ from R and S preserves positivity, equal spectra and $R - S = F$, and lowers the cost ([Mtwo], remark after (2.1)). In s -notation a triple reads

$$\sum_{i \in I, i < d} s_i - \sum_{j \in J, j > 1} s_{d+1-j} \geq \sum_{k \in K} \lambda_k. \quad (2.2)$$

With $\lambda(I) = (i_r - r, \dots, i_1 - 1)$ the partition attached to $I = (i_1 < \dots < i_r)$, (2.2) holds for all Hermitian triples whenever the Littlewood–Richardson coefficient $c_{\lambda(I)\lambda(J)}^{\lambda(K)}$ is nonzero (Klyachko [Kl98]); this necessity direction is all that the lower bounds use. Sufficiency (Klyachko, Knutson–Tao) enters only in identifying the optimal common spectra with the optimal face of (2.1) (Section 7) and inside the numerical chain LP of Section 8. We use that $\lambda \mapsto \kappa_d(\lambda)$ is convex and piecewise linear (an LP value with λ in the right-hand sides) and that $\kappa_{d+1}(F \oplus 0) \leq \kappa_d(F)$.

Forms. A form is $g(\lambda) = \sum_j \alpha_j a_j + \sum_{i < n} \beta_i b_i$ with integer coefficients, written $(\alpha_1, \dots, \alpha_m; \beta_1, \dots, \beta_{n-1})$; the coefficient of b_n is normalised to 0 using $\sum a = \sum b$. It is *valid* if $g \leq \kappa_d$ on the stratum and *exposed* if it is the unique maximiser of the set of valid forms on a nonempty open set; its *chamber* is the closed set where it is a maximiser. $S(m, n, z)$ denotes the set of exposed forms computed in Section 8.

Layerings and costs. A *layering* L is an ordered partition of the multiset $\{a_j\} \cup \{-b_i\}$, possibly together with some zeros, into layers $\Lambda_0, \Lambda_1, \dots, \Lambda_T$. Write $t(x)$ for the index of the layer of x , $t(b_i) := t(-b_i)$, $\text{tail}_u := \sum_{x \in \Lambda_u \cup \dots \cup \Lambda_T} x$, and $\text{cost}(L) := \sum_{u=1}^T \text{tail}_u$.

Lemma 2.1. $\text{cost}(L) = \sum_x t(x) x = \sum_j (t(a_j) - t(b_n)) a_j + \sum_{i < n} (t(b_n) - t(b_i)) b_i$.

Proof. $\sum_{u \geq 1} \text{tail}_u = \sum_x x \cdot \#\{u \geq 1 : u \leq t(x)\} = \sum_x t(x) x$; adding $t(b_n)(\sum_i b_i - \sum_j a_j) = 0$ gives the canonical coordinates. $\square$

Conversely a form with $\alpha_j = t(a_j) - t(b_n)$, $\beta_i = t(b_n) - t(b_i)$ for nonnegative integers t determines its *level layering* $L(g)$ (level sets of t , ordered, common shift fixed by $\min t = 0$). The *b-schedule* of a layering is $u = (u_1, \dots, u_n)$, $u_i := t(b_i)$, and the *capacity* of layer $u \geq 1$ is $c_u := \#\{i : u_i < u\}$.

Block shifts ([Mtwo] Section 5.1, valid for any layer sizes). Let $D_t := \text{diag } \Lambda_t$ and $C = \sum_{t=1}^T M_t$ with $M_t : \Lambda_{t-1} \rightarrow \Lambda_t$. With $2R_t := M_t M_t^*$ (on Λ_t) and $2S_t := M_t^* M_t$ (on Λ_{t-1}), $CC^* - C^*C = 2F$ on the layered entries iff

$$S_1 = -D_0, \quad R_t - S_{t+1} = D_t \quad (1 \leq t < T), \quad R_T = D_T, \quad R_t, S_t \text{ have the same nonzero spectrum;} \quad (2.3)$$

conversely positive semidefinite R_t, S_t with equal nonzero spectra are realised by $M_t = V\sqrt{2\Sigma}W^*$. Then $\frac{1}{2}\|C\|^2 = \sum_t \text{Tr } S_t = \text{cost}(L)$ and $\text{rank } C = \sum_t \text{rank } S_t \leq \sum_t \min(|\Lambda_{t-1}|, |\Lambda_t|)$; zeros of F left outside the layering lie in $\ker C \cap \ker C^*$. A layering is *feasible* at λ if (2.3) is solvable; then $\kappa_d \leq \text{cost}(L)$.

3. Statements

Definition 3.1 (ordered full-capacity layering $L(u)$). Let $u \in \mathbb{Z}_{\geq 0}^n$ with $\min u = 0$ and capacities $c_u = \#\{i : u_i < u\}$. Place $-b_i$ in layer u_i . Place $a_1, a_2, \dots$ in this order into layers 1, 2, ..., layer u receiving exactly c_u positives while positives remain; let T be the layer receiving the last positive (it receives q positives, $1 \leq q \leq c_T$). Require $u_i \leq T$ for all i . Write t_u for the index of the first positive in layers $\geq u$ ($t_1 = 1$, $t_{u+1} = t_u + c_u$ for $u < T$) and $B_u := \{d + 1 - i : u_i < u\}$ for the spectrum indices of the negatives before layer u ($|B_u| = c_u$).

Family A of [Mthree] is the case of nondecreasing u ; the layerings $\mathcal{L}(F_k)$ of [Mtwo] are $n = 2, u = (0, k)$; the class also contains every schedule with inversions of the negatives, e.g. $u = (0, 2, 1)$.

Theorem 1 (all $m, n \geq 1, z \geq 0$, every u as in 3.1). $\kappa_d(F) \geq \text{cost}(L(u))$ on the whole inertia- (m, n) stratum, in every dimension $d \geq m + n$. The certificate consists of Lidskii–Wielandt and Weyl triples with Littlewood–Richardson coefficient 1; each s_t with $t \leq m$ occurs exactly once with coefficient +1 and the only negative terms are $-s_\beta$ with $d - n + 1 \leq \beta \leq d - 1$.

Lemma 3.2 (rearrangement). For a fixed multiset of schedule values, $\text{cost}(L(u))$ is maximised by the nondecreasing assignment $u_1 \leq \dots \leq u_n$. Hence $\max_u \text{cost}(L(u))$ is attained by a family-A layering.

Proposition 3.3 (full capacity is necessary). For $m = n = 2, a = b = (1, 1)$, the layering $\{-b_1, -b_2\}\{a_1\}\{a_2\}$ (capacity $c_1 = 2$, one positive placed) has cost $3 > \kappa_4 = 2$.

Theorem 2A (aligned-spike cone; all m, n, z). Let $L = L(u)$ be ordered and full-capacity. The chain (2.3) has a solution with all R_t, S_t diagonal iff L carries a balanced path system (Definition 5.1); then the spectrum splits into n one-spike blocks $F_i = (a_1^{(i)} \geq a_2^{(i)} \geq \dots; -b_i)$ with $\sum_j a_j^{(i)} = b_i$, and

$$\kappa_d(F) = \text{cost}(L) = \sum_{i=1}^n \sum_j j a_j^{(i)} = \sum_i \kappa(F_i).$$

Theorem 2B (interlacing regions; all $m \geq n \geq 2, z \geq 0, 1 \leq k \leq m - n + 1$). With $L_k, \tau(k)$ and R_k as in Definition 5.5, $\kappa_d(F) = \text{cost}(L_k)$ for every $\lambda \in R_k$, where

$$\text{cost}(L_k) = k b_1 + \sum_{j \leq k} (j - k) a_j + \sum_{j > k} \left\lfloor \frac{j - k}{n} \right\rfloor a_j.$$

R_k has nonempty interior whenever $a_1 > \dots > a_m$; for $k \geq m - n + 2$ the set R_k is empty. For $n = 2$, R_k is the F_k chamber $E_{k+2} \leq b_2 \leq E_{k+1}$ of [Mtwo] Proposition 5.4(1). An optimizer of rank m is constructed explicitly.

Theorem 3 (Open question 1 of [Mtwo], first half). For a Horn triple put $\text{box}(I, J, K) := \min(|\lambda(I)|, |\lambda(J)|)$; every triple “adding $n - 1$ boxes to a rectangle” has $\text{box} \leq n - 1$. On the inertia- $(5, 4)$ stratum ($z = 0$) the exposed form $g = (3, 1, 1, 2, 2; -2, -1, 0)$ admits no certificate — no nonnegative combination of Horn inequalities of $\text{box} \leq 3$, ordering and nonnegativity constraints — while it is certified by a tiling containing one box-4 tile. Hence tiles with n boxes are necessary at $n = 4$.

4. Lower bound: proof of Theorem 1

Fix u , write $L = L(u)$ with T, t_u, B_u as in Definition 3.1, and note $B_u \subseteq \{d - n + 1, \dots, d\} = \{m + z + 1, \dots, d\}$.

Tile u for $1 \leq u < T$. Take the Lidskii–Wielandt triple $I = K = \{t_u, \dots, d\} \setminus B_u, J = \{1, \dots, r\}, r = |K| = d - t_u + 1 - c_u$. Then $\lambda(J) = \emptyset$ and $\lambda(I) = \lambda(K)$, so $c_{\lambda(I)\lambda(J)}^{\lambda(K)} = 1$, and $\sum I + \sum J = \sum K + r(r + 1)/2$ holds because $I = K$; (2.2) is the Lidskii–Wielandt inequality $\sum_{k \in K} \lambda_k(R - S) \leq \sum_{k \in K} \lambda_k(R) + \sum_{j \leq r} \lambda_j(-S)$ [Li50,Wi55], valid in every dimension. Its right side is

$$\sum_{k \in K} \lambda_k = \sum_{j \geq t_u} a_j - \sum_{i: u_i \geq u} b_i = \text{tail}_u,$$

because the positives in layers $\geq u$ are exactly $a_{t_u}, \dots, a_m$ (placed in order, all placed), zeros contribute nothing, and the negatives in layers $\geq u$ are those with index outside B_u . Its left side is $\sum_{i \in [t_u, d-1] \setminus B_u} s_i -$

$\sum_{i=d+1-r}^{d-1} s_i$ with $d+1-r = t_u + c_u$; since $t_u + c_u - 1 = t_{u+1} - 1 \leq m-1$ while $\min B_u \geq m+z+1$, the block $[t_u, t_u + c_u - 1]$ misses B_u and $B_u \setminus \{d\} \subseteq [t_u + c_u, d-1]$, so the left side equals

$$\sum_{i=t_u}^{t_u+c_u-1} s_i - \sum_{\beta \in B_u, \beta < d} s_\beta.$$

Tiles for the last layer T . Use the Weyl triples $I = K = \{j\}, J = \{1\}$ ($\lambda(J) = \emptyset, \lambda(I) = \lambda(K) = (j-1)$, coefficient 1, sum condition $j+1 = j+1$), i.e. $s_j \geq a_j$, for $j = t_T, \dots, m$. Their sum is $\sum_{j \geq t_T} s_j \geq \sum_{j \geq t_T} a_j = \text{tail}_T + \sum_{i: u_i = T} b_i \geq \text{tail}_T$.

Summation. The positive blocks $[t_u, t_{u+1} - 1]$ ($u < T$) and $[t_T, m]$ partition $\{1, \dots, m\}$, and the negative terms involve only s_β with $d-n+1 \leq \beta \leq d-1$. Adding all tiles,

$$\sum_{t \leq m} s_t - \sum_{u < T} \sum_{\beta \in B_u \setminus \{d\}} s_\beta \geq \sum_{u=1}^T \text{tail}_u = \text{cost}(L),$$

and $\sum_{t < d} s_t$ exceeds the left side by $\sum_{m < t < d} s_t + \sum_{u < T} \sum_{\beta \in B_u \setminus \{d\}} s_\beta \geq 0$. Hence every s feasible for (2.1) has $\sum_{t < d} s_t \geq \text{cost}(L)$, i.e. $\kappa_d \geq \text{cost}(L)$. Only the index bookkeeping depends on z . $\square$

Example ($n = 3, m = 4, z = 0, d = 7$, the inverted schedule $u = (0, 2, 1)$): $L = \{-b_1\}\{-b_3, a_1\}\{-b_2, a_2, a_3\}\{a_4\}$, $c_1 = 1, c_2 = 2$. Tile $u = 1$: $I = K = \{1, \dots, 6\}, J = \{1, \dots, 6\}$, reading $s_1 \geq a_1 + a_2 + a_3 + a_4 - b_2 - b_3 = \text{tail}_1$. Tile $u = 2$: $B_2 = \{5, 7\}, I = K = \{2, 3, 4, 6\}, J = \{1, 2, 3, 4\}$, reading $s_2 + s_3 - s_5 \geq a_2 + a_3 + a_4 - b_2 = \text{tail}_2$. Last layer: $s_4 \geq a_4$. Sum: $s_1 + s_2 + s_3 + s_4 - s_5 \geq \text{cost}(L) = a_2 + a_3 + 2a_4 + b_1 - b_2$, the form $(0, 1, 1, 2; 1, -1)$.

Remark 4.1 (what the proof uses). Only (i) positives in decreasing order, (ii) every non-final layer holding *exactly* c_u positives, (iii) $B_u \subseteq \{d-n+1, \dots, d\}$. The order of the negatives never enters (B_u is used as a set), which is why arbitrary b-schedules are covered. Theorem 2 of [Mthree] is the case $n = 3$, u nondecreasing; its alternative last-layer tile is unnecessary. The certificate was verified exactly by the reviewer's own code (own Fulton recursion for T_r^d , own LR counter) for every schedule of the strata $(3, 3, 0), (3, 3, 1), (4, 3, 0), (4, 4, 0), (5, 4, 0), (5, 3, 1)$ (28, 28, 43, 157, 251, 61 schedules), including the coefficient pattern stated in Theorem 1. Numerically, $\max_u \text{cost}(L(u)) - \kappa_d \leq 2 \cdot 10^{-15}$ at every one of 400 random points in each of 20 strata up to $d = 12$ (Section 8; one $2 \cdot 10^{-7}$ outlier at $(8, 4, 0)$ was a solver-tolerance artefact, re-solved to $-9 \cdot 10^{-16}$).

Proof of Lemma 3.2. The capacities c_u , hence the placement of the positives and $\sum_j t(a_j)a_j$, depend only on the multiset $\{u_i\}$. The negative part of the cost is $-\sum_i u_i b_i$ with $b_1 \geq \dots \geq b_n$, maximised by the rearrangement inequality when u_i is nondecreasing in i . $\square$

Proof of Proposition 3.3. $\text{cost} = a_1 + 2a_2 = 3$, while the four-level formula $\kappa_4 = \max(\lambda_1 - \lambda_3, \lambda_2 - \lambda_4, \lambda_1 - 2\lambda_2 - \lambda_3, \lambda_2 + 2\lambda_3 - \lambda_4)$ ([FourLevel] Theorem 3.1) gives $\max(2, 2, 0, 0) = 2$ at $(1, 1, -1, -1)$. $\square$

The failing step is visible in the proof of Theorem 1: if layer u holds $q < c_u$ positives, the next block starts at $t_u + q < t_u + c_u$, the positive blocks overlap and some s_t receives coefficient 2 (here s_2 : the tile $u = 1$ reads $s_1 + s_2 \geq a_1 + a_2$ and the Weyl tile $s_2 \geq a_2$), so the summed left side is no longer bounded by $\sum_{t < d} s_t$. Full capacity is exactly the disjointness condition $t_{u+1} = t_u + c_u$. Numerically, the maximum over under-filled ordered layerings exceeds κ_d at 50–78% of the random points in every stratum tested. Under-filled layerings *can* be valid when the schedule has inversions (families B and C of [Mthree], e.g. $\{b_1\}\{b_3, a_1\}\{b_2, a_2\}\{a_3\}$), but their certificates need Pieri tiles.

Remark 4.2 (mirror and late negatives). Applying Theorem 1 to $-F$ gives the mirror class (negatives in decreasing order at full capacity with respect to the positives before them), the relevant class when $m < n$: at $(4, 5, 0)$ none of the 62 exposed forms lies in the class of Definition 3.1, all are mirrors of $(5, 4, 0)$ forms. If some $u_i > T$, the tails beyond T are ≤ 0 and dropping their tiles proves the bound for the larger truncated cost; such schedules are dominated by moving the negative to layer T .

5. Chains and exact regions: proofs of Theorems 2A and 2B

5.1 Monomial chains are path systems

Definition 5.1. For a layering L (possibly with zeros), a *path system* is a partition of the layered entries into sequences $\pi = (x_0, x_1, \dots, x_k)$ with $x_j \in \Lambda_{u_0+j}$ (one entry per consecutive layer) such that $x_0 + \dots + x_j \leq 0$ for $j < k$ and $x_0 + \dots + x_k = 0$ (a *balanced path*).

Proposition 5.2. (i) If L admits a path system, L is feasible: $C = \bigoplus_{\pi} W_{\pi}$, W_{π} the weighted shift $e_{x_{j-1}} \mapsto w_j e_{x_j}$ with $|w_j|^2 = -2(x_0 + \dots + x_{j-1})$, satisfies $CC^* - C^*C = 2F$ on the layered entries and vanishes on the rest, and $\frac{1}{2}\|C\|^2 = \sum_{\pi} \sum_j j x_j = \text{cost}(L)$. (ii) If (2.3) has a solution with every R_t, S_t diagonal in the eigenbasis of F , then L admits a path system.

Proof. (i) On a path, $(CC^* - C^*C)e_{x_j} = (|w_j|^2 - |w_{j+1}|^2)e_{x_j}$ with $w_0 = w_{k+1} = 0$; the recursion $|w_{j+1}|^2 = |w_j|^2 - 2x_j$ has the stated solution, nonnegative exactly under the partial-sum condition, and $w_{k+1} = 0$ is the total-sum condition. Distinct paths use disjoint basis vectors. Cost: $\frac{1}{2} \sum_j |w_j|^2 = -\sum_{j \geq 1} \sum_{i < j} x_i = -\sum_i (k-i)x_i = \sum_i i x_i$ using $\sum_i x_i = 0$; by Lemma 2.1 the layering cost restricted to the path is $\sum_i (u_0 + i)x_i = \sum_i i x_i$. (ii) If R_t, S_t are diagonal, $M_t = V\sqrt{2\Sigma}W^*$ with V, W permutation matrices times phases, so every M_t and hence C is monomial; its support graph has edges only from Λ_{t-1} to Λ_t , in- and out-degree ≤ 1 and no cycles, hence is a disjoint union of directed paths through consecutive layers (isolated vertices must be zeros), and on each path the diagonal identity is the recursion of (i), which forces the partial-sum conditions. $\square$

Lemma 5.3. Let $L = L(u)$ be ordered and full-capacity (no zeros used) with a path system. Then every path starts at a negative, contains no other negative, takes exactly one positive from every layer strictly after its start up to layer $T - 1$, and possibly one from layer T . Consequently the spectrum splits into one-spike blocks $F_i = (a_1^{(i)} \geq a_2^{(i)} \geq \dots; -b_i)$ with $\sum_j a_j^{(i)} = b_i$, $a_j^{(i)}$ being the positive taken by path i in layer $u_i + j$.

Proof. A path starts with $x_0 \leq 0$, hence at a negative. In layer $u \leq T - 1$ the c_u positives lie on c_u distinct paths started at distinct negatives of layers $< u$, of which there are exactly c_u ; so every earlier negative's path is alive at u and takes a positive there. This excludes a second negative inside a path at a layer $\leq T - 1$, a path ending before layer $T - 1$, and a negative at layer T inside a path (its total would be < 0). A path ends when its sum reaches 0, i.e. when $b_i = \sum_j a_j^{(i)}$. Conversely, any assignment of the positives to alive paths with balanced spikes has partial sums ≤ 0 automatically. $\square$

Proof of Theorem 2A. The equivalence “diagonal solution iff path system” is Proposition 5.2; the block structure is Lemma 5.3. Lower bound: Theorem 1. Upper bound and cost: Proposition 5.2(i), and for path i , $\sum_j (u_i + j)a_j^{(i)} - u_i b_i = \sum_j j a_j^{(i)}$, the one-spike value of [OneSpike] Theorem 3.1. $\square$

Remarks. (a) The cone has codimension $n - 1$ (the balance equalities); on it the cost is additive over spikes whose offsets u_i make the merged layering sorted and full. (b) Alignment is a genuine restriction: $\kappa_7((10, 1, 1, 1; -13) \oplus (5; -5)) = 23 < 24 = \kappa(F_1) + \kappa(F_2)$ (author's hive LP and reviewer's Horn LP,

both 23.000000), because the spike $(5; -5)$ cannot be offset so that the merged layering is sorted and full (the block $(10, 1, 1, 1)$ ends two layers after (5)); at random two-spike direct sums additivity held at 182/200 points. (c) Numerically (all labelled numerical): 460 random aligned points with $n \leq 5$ give $|\text{cost} - \kappa_d| \leq 2.2 \cdot 10^{-15}$ (author); at 125 aligned points in five strata the reviewer's explicit C has $\|CC^* - C^*C - 2F\| \leq 6.2 \cdot 10^{-16}$ and $|\frac{1}{2}\|C\|^2 - \kappa_d| \leq 6.7 \cdot 10^{-16}$.

5.2 One rank-one interlacing step

Lemma 5.4 (rank-one update, both directions). Let $D = \text{diag}(d_1 \geq \dots \geq d_N)$ and $s \geq 0$. The spectra of $D + s vv^*$ over unit vectors v are exactly the vectors $\sigma_1 \geq \dots \geq \sigma_N$ with

$$\sigma_1 \geq d_1 \geq \sigma_2 \geq d_2 \geq \dots \geq d_{N-1} \geq \sigma_N \geq d_N, \quad \sum \sigma = \sum d + s.$$

Proof. Necessity is Cauchy–Weyl interlacing for a positive semidefinite rank-one perturbation plus the trace. Sufficiency for distinct d_i : with $v_i^2 := \prod_j (\sigma_j - d_i) / (s \prod_{j \neq i} (d_j - d_i))$, interlacing gives numerator and denominator the sign $(-1)^{N-i}$, so $v_i^2 \geq 0$; the secular identity $\det(D + s vv^* - x) = \prod_i (d_i - x) (1 + s \sum_i v_i^2 / (d_i - x))$ then has exactly the roots σ_j , and $\sum_i v_i^2 = 1$ follows from the trace condition [Go73, Chu98, HJ]. Tied d_i : the attainable set (continuous image of the sphere) and the interlacing polytope are closed and vary continuously with d ; take limits. $\square$

Definition 5.5. For $n \geq 2$ and $1 \leq k \leq m$ let

$$L_k := \{-b_1\}, \{a_1\}, \dots, \{a_{k-1}\}, \{a_k, -b_2, \dots, -b_n\}, \{a_{k+1}, \dots, a_{k+n}\}, \{a_{k+n+1}, \dots, a_{k+2n}\}, \dots$$

(last layer partial), the ordered full-capacity layering with schedule $u = (0, k, \dots, k)$, and $\tau_i(k) := \sum_{r \geq 0} a_{k+i+rn}$ ($i = 1, \dots, n$; $a_j := 0$ for $j > m$), the column sums of the pure phase, $\tau_1 \geq \dots \geq \tau_n \geq 0$, $\sum_i \tau_i = A_{k+1}$. The tails are $A_u - B'$ ($u \leq k$) and A_{k+1+rn} ($r \geq 0$), which gives the form of Theorem 2B. Let

$$R_k := \{\lambda \text{ in the stratum} : \tau_i(k) \leq b_i \leq \tau_{i-1}(k) \text{ for } i = 2, \dots, n\}.$$

Proof of Theorem 2B. Range. If $k \geq m - n + 2$ then $k + n - 1 > m$, so $\tau_{n-1}(k) = 0$ and R_k would force $b_n \leq 0$; R_k is empty. If $k \leq m - n + 1$ and $a_1 > \dots > a_m$, then $\tau_{i-1}(k) - \tau_i(k) \geq a_{k+i-1} - a_{k+i} > 0$ for $i = 2, \dots, n$ (with $a_{m+1} = 0$), so the box $\{\tau_i < b_i < \tau_{i-1}\}$ is open and nonempty; at its lower corner $b_i = \tau_i$ ($i \geq 2$) one has $b_1 = P - \sum_{i \geq 2} \tau_i = a_1 + \dots + a_k + \tau_1 > \tau_1 \geq b_2$, so the box meets the stratum in an open set.

Lower bound. L_k is ordered and full-capacity ($c_u = 1$ for $u \leq k$, $c_u = n$ for $u > k$), so Theorem 1 applies.

Singleton phase. $S_1 = b_1$ and, layer by layer, $R_u = S_u$ (scalars) and $S_{u+1} = S_u - a_u$, so $S_u = b_1 - a_1 - \dots - a_{u-1} = A_u - B'$ for $u \leq k$. These are nonnegative on R_k : $b_i \leq \tau_{i-1}$ gives $B' \leq \tau_1 + \dots + \tau_{n-1} \leq A_{k+1} \leq A_k$ (strictly positive inside the stratum).

Mixed layer k . $S_k = s := A_k - B'$ is 1×1 ; take $R_k = s vv^*$ on Λ_k with v a unit vector of $\mathbb{C}^n$ (same nonzero spectrum). Then $S_{k+1} = R_k - D_k = s vv^* + \text{diag}(-a_k, b_2, \dots, b_n)$. By Lemma 5.4 with $d = (b_2 \geq \dots \geq b_n \geq -a_k)$, every σ with $\sigma_1 \geq b_2 \geq \sigma_2 \geq b_3 \geq \dots \geq b_n \geq \sigma_n \geq -a_k$ and $\sum \sigma = B' - a_k + s = A_{k+1}$ is attainable. Choose $\sigma = \tau(k)$: the trace matches, $\tau_n \geq 0 \geq -a_k$, and the remaining conditions are $\tau_i \leq b_i \leq \tau_{i-1}$ ($i = 2, \dots, n$), i.e. $\lambda \in R_k$. So S_{k+1} is positive semidefinite with spectrum τ .

Pure phase (diagonal continuation). For $r \geq 0$ put $\tau_i^{(r)} := \sum_{r' \geq r} a_{k+i+r'n}$ and $R_{k+1+r} := \text{diag}(\tau_1^{(r)}, \dots, \tau_n^{(r)})$ on the slots $a_{k+1+rn}, \dots, a_{k+n+rn}$ of Λ_{k+1+r} (exhausted columns carry 0; a partial

layer uses only its existing slots, and the nonzero spectra still match). Then $\text{spec } R_{k+1} = \tau = \text{spec } S_{k+1}$; $S_{k+2+r} = R_{k+1+r} - D_{k+1+r} = \text{diag}(\tau^{(r+1)})$ is positive semidefinite with the nonzero spectrum of R_{k+2+r} ; and at the last layer $\tau_i^{(r)} = a_{k+i+rn}$, so $R_T = D_T$. Thus (2.3) holds and $\kappa_d \leq \text{cost}(L_k)$ on R_k .

Rank. $\text{rank } C = \sum_t \text{rank } S_t = k + \sum_{r \geq 0} \#\{i : k + i + rn \leq m\} = k + (m - k) = m$.

$n = 2$. $\tau_1(k) = E_{k+1}$, $\tau_2(k) = E_{k+2}$ in the notation of [Mtwo], so $R_k = \{E_{k+2} \leq b_2 \leq E_{k+1}\}$, the region of Proposition 5.4(1) there. $\square$

Remarks. (a) The chain is diagonal except for one rank-one interlacing step, and its rank equals the inertia bound $\max(n_+, n_-)$ when $m \geq n$. (b) Numerical: 1 320 random points of the regions R_k in nine strata up to $(6, 4, 1)$ and $(5, 5, 0)$ give $\max |\text{cost}(L_k) - \kappa_d| \leq 2.2 \cdot 10^{-15}$ (author); the reviewer's explicit C (Golub's v , eigendecomposition of S_{k+1} , diagonal continuation) at 10 points of every nonempty R_k in six strata up to $(7, 4, 0)$ has residual $\|CC^* - C^*C - 2F\| \leq 1.6 \cdot 10^{-15}$, $|\text{cost} - \kappa_d| \leq 1.1 \cdot 10^{-15}$ and rank m at every point, and reports R_k empty exactly for $k > m - n + 1$. (c) For $n \geq 3$, R_k is in general a proper subset of the chamber of $\text{cost}(L_k)$: with b_n pushed 5% below τ_n , L_k remains optimal at 37/44 points for $(5, 3, 0)$ and 37/42 for $(6, 4, 1)$; at $(6, 3)$ the union $\cup_k R_k$ covers 67 of the 92 points (of 400) where some L_k is an argmax, at $(7, 4)$ 13 of 35, while at $(4, 2)$, $(6, 2)$, $(4, 3)$, $(5, 4)$, $(6, 5)$ the two sets coincide on all samples. The missing part of the chamber needs non-diagonal steps in the pure phase (for $n = 2$, the 2×2 rotations of [Mtwo] Lemma 5.2). (d) The same mechanism gives sufficient regions for every family-A schedule with several mixed layers (at layer u_i one alive path absorbs the new negatives by a rank-one step, the others continue diagonally), a finite union of polytopes; this general version was neither written out nor tested and is not claimed. For schedules with several negatives in Λ_0 (e.g. $u = 0^n$) the initial state is forced and the pure-phase freedom is essential.

5.3 The pure phase: a necessary condition

Consider a state $\sigma = \text{spec } S$ (n values) followed by pure layers $\delta^{(1)}, \dots, \delta^{(q)}$ of n sorted positives each (the last possibly shorter), with column sums $\tau_i = \sum_t \delta_i^{(t)}$.

Proposition 5.6. If the pure chain is feasible from σ , then $\sigma < \tau$ (majorisation: the bottom- j sums of σ dominate those of τ for every j).

Proof. By Lidskii, $\text{spec}(R_t - D_t) - \text{spec } R_t$ lies in the permutohedron $\Pi(-\delta^{(t)})$. Summing along the chain, $\sigma^{(q)} - \sigma \in \sum_t \Pi(-\delta^{(t)}) = \Pi(-\sum_{t < q} \delta^{(t)})$ (Minkowski sums of permutohedra of similarly sorted vectors are permutohedra of the sums, since the support functions $c \mapsto \langle c^\downarrow, x^\downarrow \rangle$ add), with $\sigma^{(q)} = \delta^{(q)}$ sorted. Hence $\sigma \in \delta^{(q)} + \Pi(\tau') \subseteq \Pi(\tau)$, $\tau' = \sum_{t < q} \delta^{(t)}$. $\square$

Numerically (chain LP of Section 8 in dimension n): for $n = 2$ the exact region is $\tau_2 \leq \sigma_2 \leq \tau_1 - \max_t(\delta_1^{(t)} - \delta_2^{(t)})$ (no violation either way at 140 feasible of 800 sampled points; it is the F_0 region $E_2 \leq b_2 \leq E_1 - g^*$ of [Mtwo]); for $n = 3, 4$ majorisation is never violated (0 of 67 feasible points) but the analogous gap inequalities are (13–14 of 24–27 at $n = 3$), so the exact region for $n \geq 3$ lies strictly between the two and its facets are not identified — the obstacle to a closed form for the F_0 -type chambers.

6. Which Horn triples are needed: Theorem 3 and data

Open question 1 of [Mtwo] (Section 10 there) reads verbatim: “*Inertia $(m, 3)$ and beyond.* The certificate here uses one Pieri-type triple per even index; a natural guess is that inertia (m, n) requires triples adding $n - 1$ boxes to a rectangle, and that block shifts with blocks of size up to n attain. Neither

is known; the $d = 8$ example of [RankOnset] with $r_* > r_0$ has inertia $(4, 4)$ and shows that the rank- m block picture cannot persist unchanged.” The Pieri triple of [Mtwo] Lemma 4.3 has $\lambda(J) = (1)$ and $\lambda(K)$ a rectangle; the natural reading of the first half is that tiles of that type with $|\lambda(J)| = n - 1$ suffice. Every such tile has box $\leq n - 1$, so the Horn LP restricted to box $\leq n - 1$ upper-bounds everything the guess can certify.

Proof of Theorem 3. Let $H_3(\lambda)$ be the value of (2.1) with the Horn constraints restricted to the box- ≤ 3 triples of T_r^9 (box 0 is the Lidskii-Wielandt/Weyl class: $I = K, J = (1..r)$ or $J = K, I = (1..r)$). Suppose g had a certificate from this class: nonnegative multipliers y on the box- ≤ 3 inequalities, w on the ordering constraints $s_t \geq s_{t+1}$ and v on $s_t \geq 0$, with $\sum_k y_k A_k + \sum_t w_t (e_t - e_{t+1}) + v = \mathbf{1}$ and $\sum_k y_k B_k = g$ as linear forms in λ . Then every s feasible for the restricted program satisfies $\sum_{t < d} s_t \geq g(\lambda)$, i.e. $H_3(\lambda) \geq g(\lambda)$ for every λ . At the rational point

$$a = \left(\frac{97}{127}, \frac{163}{979}, \frac{86}{1945}, \frac{29}{1445}, \frac{1}{184} \right), \quad (b_2, b_3, b_4) = \left(\frac{241}{885}, \frac{479}{1954}, \frac{209}{1774} \right), \quad b_1 = \sum_j a_j - b_2 - b_3 - b_4$$

(denominators ≤ 2000 except for b_1 , which restores the trace), the vector

$$s = \left(\frac{97}{127}, \frac{34142468748934736}{84409551000378285}, \frac{479}{1954}, \frac{209}{1774}, \frac{3891}{265880}, 0, 0, 0 \right)$$

is nonincreasing, nonnegative, and satisfies every box- ≤ 3 inequality of T_r^9 (13 328 of the 39 716 triples of T_r^9 have box ≤ 3) in exact rational arithmetic, with

$$\sum_t s_t = \frac{10550738857556475461}{6825198573622282920} \approx 1.545851 < g(\lambda) = \frac{57542567000354849}{37093470508816755} \approx 1.551286.$$

Hence $H_3(\lambda) \leq \sum_t s_t < g(\lambda)$, a contradiction. The tiling certificate of g is: the Weyl tiles $s_1 \geq a_1$ ($I = J = K = \{1\}$), $s_3 \geq b_3$ ($I = (1..8), J = K = (1..9) \setminus \{7\}$) and $s_4 \geq b_4$ ($I = (1..8), J = K = (1..9) \setminus \{6\}$), together with

$$I = (1, 2, 5, 6), \quad J = (1, 2, 3, 9), \quad K = (1, 4, 5, 9) : \quad s_2 + s_5 + s_6 - s_7 - s_8 \geq a_1 + a_4 + a_5 - b_1,$$

with $\lambda(I) = (2, 2)$, $\lambda(J) = (5)$, $\lambda(K) = (5, 2, 2)$, sum condition $29 = 19 + 10$, membership in T_4^9 , and $c_{(2,2)(5)}^{(5,2,2)} = 1$ by the Pieri rule [Fu97] ($((5, 2, 2)/(2, 2))$ is a horizontal strip); its box is 4. The sum is $s_1 + \dots + s_6 - s_7 - s_8 \geq 2a_1 + a_4 + a_5 - b_1 + b_3 + b_4 = 3a_1 + a_2 + a_3 + 2a_4 + 2a_5 - 2b_1 - b_2 = g$, each s_t with $t \leq 5$ once. $\square$

At the (floating-point) point above, $\kappa_9 = g(\lambda) = 1.551286$ with margin 0.069 over the next form of $S(5, 4, 0)$, and the restricted values for $B = 0, \dots, 5$ are 1.531217, 1.531217, 1.545850, 1.545850, 1.551286, 1.551286: the jump to κ occurs exactly at $B = 4$. The certificate is the dual returned by both the author’s and the reviewer’s LP (a first transcription listed $s_2 \geq b_2$ instead of $s_3 \geq b_3, s_4 \geq b_4$, which is not a tiling since the box-4 tile already contains s_2). The exact computation was carried out by the reviewer and is reproduced, with the rational point printed, by `repro/exact_oq1.py`; the multipliers of a general certificate need not be 1, and the weak-duality argument covers them all.

Box counts on the computed strata (numerical). For λ in the stratum let $B_{\min}(\lambda)$ be the least B such that the box- $\leq B$ program attains $\kappa_d(\lambda)$ (floating-point HiGHS, tolerance 10^{-9} , gaps $\geq 10^{-3}$ whenever a level is reported insufficient), and for an exposed form g let $B_{\min}(g)$ be $B_{\min}$ at a point interior to both its chamber and the stratum (there a restricted certificate attaining κ certifies g).

(m, n, z)	$ S $	$B_{\min}(g)$ histogram	max
$(m, 2, z), d \leq 9$	—	(random points only)	1
$(3, 3, 0), (3, 3, 1)$	10	0:6, 1:4	1
$(4, 3, 0) / (4, 3, 1)$	19 / 18	0:10, 1:5, 2:4 / 2:3	2
$(5, 3, 0) / (5, 3, 1)$	27 / 26	0:13, 1:9, 2:3, 3:2 / 3:1	3
$(6, 3, 0)$	34	0:17, 1:13, 2:3, 3:1	3
$(4, 4, 0) / (4, 4, 1)$	32	0:14, 1:8, 2:10	2
$(5, 4, 0) / (4, 5, 0)$	63 / 62	0:21, 1:13, 2:19, 3:8, 4:2 / 3:7, 4:2	4

So the box count is $\leq n$ in all data and reaches n for $n = 3$ ($m \geq 5$; the reviewer reproduced the $(5, 3, 0)$ histogram with own chamber points and own LP, the two box-3 forms being $(1, 2, 3, 3, 4; -2, -1)$ and $(3, 1, 2, 2, 3; -2, -1)$) and for $n = 4$ ($m = 5$). The tile list of [Mthree] used $\lambda(J) = (3)$ tiles; that box 3 is necessary at $n = 3$ is new here (numerical). The forms needing n boxes are b-inverted layerings with the largest negative entering late, e.g. $(3, 1, 1, 2, 2; -2, -1, 0) = \{b_3, b_4\}\{b_2, a_2, a_3\}\{b_1, a_4, a_5\}\{a_1\}$ and $(1, 1, 2, 3, 4; -2, -1, 0) = \{b_3, b_4\}\{b_2, a_1, a_2\}\{b_1, a_3\}\{a_4\}\{a_5\}$. *Rectangles*: among the active non-Lidskii tiles of the full-LP duals, $\lambda(K)$ is a rectangle at only $1/38$ ($(5, 4, 0)$), $0/32$ ($(4, 4, 1)$), $3/31$ ($(4, 5, 0)$), $5/27$ ($(6, 3, 0)$) occurrences; the rectangle picture of the $(m, 2)$ Pieri triple does not persist. *Shapes*: the most frequent non-Lidskii shapes are Pieri rows and columns ($\lambda(J) = (3)$ with a horizontal strip at $n = 3$; $(6), (4), (2), (1, 1), (3, 3)$ at $n = 4$).

Pieri shapes do not always suffice. A first draft conjectured that every exposed form has a tiling whose non-Lidskii tiles are Pieri triples ($\lambda(I)$ or $\lambda(J)$ a single row or column). The reviewer tested the Horn LP restricted to box-0 tiles plus *all* triples in which $\lambda(I)$ or $\lambda(J)$ is a single row or column of any length (12 798 of the 39 716 triples at $d = 9$; a superset of the conjectured class, with arbitrary multipliers) at interior chamber points of every exposed form: all forms of $(3, 3, 0), (4, 3, 0), (5, 3, 0), (4, 4, 0), (4, 3, 1), (5, 3, 1), (4, 4, 1)$ pass, and exactly one form of $(5, 4, 0)$ fails, $g' = (1, 1, 2, 3, 4; -2, -1, 0)$, with gaps 0.008, 0.013, 0.016 at three chamber points. The obstruction is exact as well (repro/exact_oq1.py): at the rational point $a = (\frac{409}{1715}, \frac{179}{878}, \frac{283}{1424}, \frac{339}{1784}, \frac{203}{1202})$, $(b_2, b_3, b_4) = (\frac{325}{1224}, \frac{495}{1963}, \frac{206}{955})$, $b_1 = \sum_j a_j - b_2 - b_3 - b_4$, the vector $s = (\frac{384815}{1072184}, \frac{325}{1224}, \frac{495}{1963}, \frac{206}{955}, \frac{203}{1202}, \frac{46058}{3366545}, 0, 0)$ satisfies every inequality of that class exactly, with $\sum_t s_t = \frac{67237875494435837}{52741022880857220} \approx 1.274869 < g'(\lambda) = \frac{29791432143128927309}{23153309044696319580} \approx 1.286703$; the same value is reached by an exact box- ≤ 3 vertex, so g' also needs box 4. Its returned certificate (every tile in T_r^9 with LR coefficient 1 by both implementations) is

tile $(I; J; K)$	s-form $\geq$ right side	$\lambda(I), \lambda(J), \lambda(K)$
$(5; 1; 5)$	$s_5 \geq a_5$	Weyl
$(1, 5; 1, 5; 4, 5)$	$s_1 \geq a_4 + a_5$	$(3), (3), (3, 3)$
$(1, 2, 5, 6; 1, 2, 5, 9; 3, 4, 5, 9)$	$s_2 + s_6 - s_8 \geq a_3 + a_4 + a_5 - b_1$	$(2, 2), (5, 2), (5, 2, 2, 2)$
$(1..8; [9] \setminus 7; [9] \setminus 7)$	$s_3 \geq b_3$	Weyl
$(1..8; [9] \setminus 6; [9] \setminus 6)$	$s_4 \geq b_4$	Weyl

The third tile has a two-row $\lambda(J)$, and the sum is $s_1 + \dots + s_6 - s_8 \geq a_1 + a_2 + 2a_3 + 3a_4 + 4a_5 - 2b_1 - b_2 = g'$. The form g' is a $z = 0$ -only form (it violates the inversion balance of Section 8 and is not exposed at

$z = 1$). So the conjecture of Section 8.3 is stated with a box bound only, and the shape statement is Question 8.4.

7. The $d = 8$ example

The spectrum $\lambda = (5, 1, 1, 1, -2, -2, -2, -2)$ of [RankOnset] Theorem 4.1 ($4c - 3b, b, b, b, -c^4$ with $b = 1, c = 2$) has $\kappa_8 = 13$ (hive and Horn LP, both authors' codes) and $r_* = 5 > 4 = r_0$. Exactly two of the 32 exposed forms of $S(4, 4, 0)$ attain 13 there:

$$(3, 1, 1, 2; -2, -1, 0) = \{b_3, b_4\}\{b_2, a_2, a_3\}\{b_1, a_4\}\{a_1\}, \quad (4, 1, 2, 2; -3, -2, -1) = \{b_4\}\{b_3, a_2\}\{b_2, a_3, a_4\}\{b_1\}\{a_1\}.$$

Both level layerings are chain-feasible at the point (blocks of size ≤ 3), and both have chain-rank bound $\sum_t \min(|\Lambda_{t-1}|, |\Lambda_t|) = 5$. On the optimal face of the Horn LP, $\min s_5 = 1$ exactly (integer data; HiGHS returns $s = (5, 3, 2, 2, 1, 0, 0)$, and $\min s_6 = \min s_7 = 0$), so every optimizer has rank 5, as [RankOnset] proved — and the rank-5 optimizers *are* block shifts. The extra rank comes from b_1 entering a late layer ($\{b_1, a_4\}$ or $\{b_1\}$ after the big layer), which raises the chain-rank bound to 5; there is no padding ($z = 0$). The point is maximally degenerate (the LP duals are not unique: the hive LP returns a non-integer subgradient, the reviewer's Horn LP an integer one outside $S(4, 4, 0)$), which is why the two exposed forms had to be identified through the computed set. Along the family $c \in [0.8, 3]$, $b = 1$: $r_* = 5$ for $c \geq 1.1$ with $\operatorname{argmax} (3, 1, 1, 2; -2, -1, 0)$ (or $(4, 1, 2, 2; -3, -2, -1)$ at $c = 3$); $r_* = 4$ at $c = 1$ (all entries ± 1); $r_* = 5$ again for $c < 1$, where the sorted spectrum changes, with $\operatorname{argmax} (2, 3, 3, 1; -2, -2, -1)$. This sharpens the remark in Open question 1: the block picture persists at $d = 8$, but not the rank- m block picture — the relevant layerings have the largest negative in a late layer, inverted schedules that Theorem 1 covers as lower bounds but that the families of [Mtwo, Mthree] do not contain for $n = 4$.

8. Numerical structure and a conjecture for inertia (m, n)

Everything in this section is NUMERICAL (floating-point HiGHS; hive LP of [Mthree] and explicit Horn LP over Fulton's T_r^d for $d \leq 9$, agreeing to $\leq 4 \cdot 10^{-15}$ at every point used). Sampling laws: exponential, uniform, squared normal, tie-heavy (integer entries 1–3), dominant a_1 with flat b ; $P = 1$.

8.1 Exposed forms. Hive-LP dual gradients at 1 500–8 000 random points were rationalised; every integer form found was checked for validity (joint LP) and exposedness (radius LP), and the maximum of the resulting set equals κ_d at 300 fresh points per stratum to $\leq 4 \cdot 10^{-15}$. The $(m, 3)$ counts reproduce $S(m, z)$ of [Mthree]; the reviewer's own Horn-LP gradients reproduce the sets at $(3, 3, 0)$ and $(4, 4, 0)$ exactly. In the table, “ordered” is the number of forms in the class of Definition 3.1 (for $(4, 5, 0)$: 0, and 7 in the mirror class), “ a -ord.” the number whose level layering has the positives in decreasing order, and “ $\kappa = \text{OL}$ ” the share of 400 random points (300 for $d \leq 7$) at which κ_d equals the ordered bound $\max_u \operatorname{cost}(L(u))$ (in every run equal to the family-A maximum, as Lemma 3.2 predicts; the dominant- a_1 law never lies in this region).

(m, n, z)	d	$ S $	ordered	a -ord.	max layer	$\kappa = \text{OL}$ (%)
$(3, 3, 0) /$ $(3, 3, 1)$	6 / 7	10 / 10	2	4	3	22.0 / 19.7
$(4, 3, 0) /$ $(4, 3, 1)$	7 / 8	19 / 18	5	9 / 8	3	38.7 / 32.2

(m, n, z)	d	$ S $	ordered	a -ord.	max layer	$\kappa = \text{OL} (\%)$
$(5, 3, 0) / (5, 3, 1)$	8 / 9	27 / 26	8	13 / 12	3	43.5 / —
$(6, 3, 0) / (6, 3, 1)$	9 / 10	34 / —	11	14	3	— / 53.8
$(4, 4, 0) / (4, 4, 1)$	8 / 9	32 / 32	3	8	4	12.8 / 11.8
$(5, 4, 0) / (5, 4, 1)$	9 / 10	63 / 62	7	19	4	24.5 / 26.8
$(4, 5, 0)$	9	62	0	15	4	—
$(6, 4, 0) / (6, 4, 1)$	10 / 11	92 / —	12	24	4	30.2 / 33.0
$(7, 4, 0) / (8, 4, 0)$	11 / 12	—	—	—	—	43.2 / 48.8
$(5, 5, 0) / (5, 5, 1)$	10 / 11	108 / —	4	19	5	7.8 / 7.0
$(6, 5, 0)$	11	—	—	—	—	18.5

The ordered region grows with m at fixed n and shrinks with n . Every exposed form's level layering has Λ_0 purely negative, Λ_T purely positive, no empty layer, at most c_u positives in layer u , and layers of size $\leq \min(m, n)$ (100% of 10–108 forms in all 14 strata; reviewer: 0 violations in all 15 files). *Padding*: $S(4, 4, 0) = S(4, 4, 1)$; $S(5, 4, 0) \neq S(5, 4, 1)$ with three $z = 0$ -only forms, $(0, 1, 1, 2, 3; -1, 0, 1)$, $(1, 1, 2, 3, 4; -2, -1, 0)$, $(2, 1, 2, 3, 4; -2, -1, -1)$, and two $z = 1$ -only forms, $(-1, 0, 0, 1, 2; 1, 2, -1)$, $(1, 2, 2, 3, 4; -2, -1, -1)$ (60 common) — the first occurrence of exposed forms *created* by padding (for $n = 3$ padding only removes forms); $(4, 3)$ and $(5, 3)$ have exactly one $z = 0$ -only form each, $(6, 4, 0)$ has five forms violating the inversion balance below and $(5, 5, 0)$ none.

8.2 Blocks of size $\min(m, n)$ attain (numerical). The chain LP of [Mthree] Section 5.2 (Horn program in dimension ≤ 5 per step, rank caps, ordering, trace and terminal conditions) was ported to blocks of size ≤ 5 ; at each of 300 random points per stratum every exposed form tied at κ_d (tolerance 10^{-8}) was tested through its level layering, with up to two inserted zeros when $z \geq 1$. In all 13 strata $((3, 3, 0), (3, 3, 2), (4, 3, 0), (5, 3, 0), (6, 3, 0), (5, 3, 1), (4, 4, 0), (4, 4, 1), (5, 4, 0), (5, 4, 1), (4, 5, 0), (6, 4, 0), (5, 5, 0))$ some tied form's level layering is chain-feasible at 300/300 points, with blocks of size $\leq \min(m, n)$ and no inserted zero except at one point each of $(5, 3, 1)$ and $(5, 4, 1)$. Testing only the dual form returned by the LP fails at tie-heavy points where that form's layering is provably never chain-feasible (e.g. $(1, 1, 2, 1, 2; -1, 1, -1)$ at $(5, 4, 0)$), while another tied form works. The reviewer did not re-implement the chain LP; it is the author's port of the cycle-2 code audited then.

8.3 Conjecture (inertia (m, n)). For every (m, n) and z , κ_d is the maximum of a finite set $S(m, n, z)$ of integer forms, each the cost of its level layering $L(g)$, and:

1. (*structure*) Λ_0 consists of negatives only, Λ_T of positives only, no layer is empty; layer $u \geq 1$ holds at most c_u positives (capacity rule) and $|\Lambda_u| \leq \min(n_+, n_-)$ for all u ;
2. (*inversion balance*) for $z \geq 1$, $\#\{j < j' : t(a_j) > t(a_{j'})\} + \sum_u (c_u - q_u) \cdot \#\{\text{positives in layers} > u\} = \#\{i < i' : t(b_i) > t(b_{i'})\}$, q_u the number of positives in layer u (the n -negative version of the rule of [Mthree]); the forms violating it are exactly the $z = 0$ -only forms;
3. (*ordered part*) the exposed family-A forms are those whose chamber meets the interior of the stratum, and the mirror family plays the same role for $n > m$;

4. (*upper bound*) for every $g \in S(m, n, z)$ and every λ in its chamber, $L(g)$, with at most one inserted zero when $z \geq 1$, is chain-feasible; so κ_d is attained by block weighted shifts with blocks of size $\leq \min(n_+, n_-)$ in every dimension, while the minimal optimal rank can exceed $\max(n_+, n_-)$ already at $z = 0$ (Section 7);
5. (*lower bound*) every $g \in S(m, n, z)$ has a tiling certificate (multipliers 1, each s_t with $t \leq m$ once) whose tiles have box $\leq \min(n_+, n_-)$, and for some forms no certificate with fewer boxes exists once $m \geq n + 1$ (proved for $(5, 4)$: Theorem 3);
6. (*padding*) $S(m, n, z)$ is independent of $z \geq 1$; $S(m, n, 0) \setminus S(m, n, 1)$ consists of the forms violating 2, and their existence is what gives $\kappa_d > \kappa_{d+1}(\lambda \oplus 0)$ on an open set (the $z = 1$ -only forms are dominated at $z = 0$, e.g. $(2, 1, 2, 3, 4; -2, -1, -1) - (1, 2, 2, 3, 4; -2, -1, -1) = a_1 - a_2 \geq 0$); a padding phase exists for $(m, n) = (5, 4), (4, 5), (6, 4)$ but not for $(4, 4)$ or $(5, 5)$, so the set of (m, n) with a padding phase is not a congruence condition in m alone ($n = 3: 3 \nmid m$).

Numerical status: 1 holds for all 14 strata; 2 holds for all forms at $z \geq 1$ $((3, 3, 1), (4, 3, 1), (5, 3, 1), (4, 4, 1), (5, 4, 1))$ and fails at $z = 0$ exactly for the $z = 0$ -only forms of $(4, 3), (5, 3)$ and for 3 forms of $(5, 4, 0)$, 3 of $(4, 5, 0)$, 5 of $(6, 4, 0)$; 4: 300/300 in 13 strata; 5: box histograms of Section 6. What is proved towards the conjecture: Theorem 1 (every ordered form is valid, all d), Lemma 3.2, Theorem 2A, Theorem 2B, Proposition 5.6, Theorem 3.

8.4 Question (Pieri shapes). At $z \geq 1$ every exposed form of the seven strata tested has a certificate built from Lidskii–Wielandt tiles and tiles with $\lambda(I)$ or $\lambda(J)$ a single row or column, whereas the $z = 0$ -only form $g' = (1, 1, 2, 3, 4; -2, -1, 0) \in S(5, 4, 0)$ has none (Section 6). Do all forms of $S(m, n, z)$ with $z \geq 1$ — equivalently, the forms that survive padding — admit tilings with Pieri-shaped tiles of at most $\min(n_+, n_-)$ boxes? The general- z validity of such tilings would follow from the strip condition of the Pieri rule tile by tile, as sketched in [Mthree] Section 4.3.

9. Open questions

1. *The chambers of L_k for $n \geq 3$.* Describe the chain-feasibility region of L_k beyond R_k (non-diagonal pure phase) and the facets of the exact pure-phase region of Section 5.3; for F_0 -type layerings ($u = 0^n$) this is the whole problem.
2. *Several mixed layers.* Write out and test the multi-step interlacing regions of Remark (d) in Section 5.2, which would give closed forms for every family-A layering on a union of polytopes.
3. *Beyond the ordered class.* The b-inverted exposed forms need Pieri or two-row tiles; a general- z proof of their validity and a characterisation of the exposed layerings are open. Question 8.4 is the shape part.
4. *Padding phases.* Prove the inversion-balance rule, determine the set of (m, n) with a padding phase, and explain why padding can create exposed forms for $n \geq 4$.
5. *Optimizers.* Classify the chambers by minimal optimal rank (Section 7: rank $m + 1$ at $z = 0$ for $(4, 4)$) and by uniqueness of the optimal spectrum.
6. *Growth.* A formula, or the growth rate, for $|S(m, n, z)|$ (10, 19, 27, 34 at $n = 3$; 32, 63, 92 at $n = 4$; 108 at $(5, 5)$).

10. Verification and reproducibility

Author's code (Python 3.12, numpy/scipy with HiGHS; repro/author/): `mn_tools.py`, `joint_lp.py` (validity and radius LPs), `horn_lp.py` (Horn LP over T_r^d with box filter, $d \leq 9$), `hive_core.py`, `ol_test.py` (Theorem 1 and Lemma 3.2 numerically, 20 strata), `interlace_test.py`, `region_coverage.py`

(Theorems 2A, 2B), `pure_chain.py` (Section 5.3), `forms_mn.py`, `inv_rule.py` (exposed forms and rules; JSON files `forms_m{m}_n{n}_z{z}.json`), `horn_boxes.py`, `form_boxes.py` (box counts and shapes), `chain_lp.py` (Section 8.2), `d8_example.py` (Section 7), with logs.

Reviewer’s independent code (`repro/reviewer/`): `myhorn.py` — own T_r^d by the Horn–Fulton recursion (counts 41, 142, 522, 2062, 8752, 39716 for $d = 4, \dots, 9$, set-equal to the author’s lists and to the LR-positivity criterion for $d \leq 7$), own LR counter, own Horn LP with box filter, duals and optimal-face minimisation, validated against the four-level and one-spike formulas and the hive LP (10^{-15}); `test_thm1.py` (exact certificate check of Theorem 1 for all 568 schedules of six strata; Proposition 3.3), `test_thm2AB.py` (explicit weighted shifts and interlacing chains, Section 5), `test_oq1.py` (Theorem 3, the $(5, 3, 0)$ histogram, the exposed sets at $(3, 3, 0)$ and $(4, 4, 0)$, the $d = 8$ example). The reviewer also re-derived every proof.

Exact arithmetic is used in the certificate of Theorem 1 (integer identities, LR coefficients), in the rational points and vertices of Section 6 (`repro/exact_oq1.py`, which also verifies both certificates there: membership in T_r^9 , LR coefficients by two implementations, integer tiling identities), and in the $d = 8$ optimal face. Everything else is floating point with the tolerances stated.

Package `repro/`: `author/` (verbatim working directory plus the four Fulton pickles it reads, one marked path line), `reviewer/` (two absolute paths made package-relative, marked `REPRO`), `exact_oq1.py`, `run_fast.sh` (reruns the fast scripts into `logs/`, times in `logs/timings.txt`; `forms_mn.py` is not rerun because it overwrites the data files) and `README.md` with commands, runtimes and expected final lines. Literature queries (2026-09-14, three WebSearch queries on the prescribed-spectrum self-commutator cost and Horn certificates, on the rank-one interlacing converse, and on Lidskii certificates with block bidiagonal weighted shifts) returned only commutator norm inequalities, almost-commuting matrices, limiting Horn inequalities (arXiv:2410.08907), the rank-one sources [Go73,Chu98] and [Fu00].

AI-assistance statement

The mathematics, computations and text of this manuscript were produced by Claude Fable 5.1 agents (Anthropic) working under the direction of the author, who set the problem, selected the line of records and is responsible for the claims. One agent carried out the research (proofs, scripts, experiments) and wrote the report on which this manuscript is based; a second, independent agent performed an adversarial review with its own implementations of every computational ingredient, verified the constructions, upgraded the refutation of Open question 1 to exact arithmetic, and found the corrections incorporated here (range of k in Theorem 2B, the Weyl tiles of the box-4 certificate, the counterexample to the Pieri-shape statement, a mis-citation and a redundant sentence); a third agent wrote this text, added `exact_oq1.py`, assembled the reproducibility package and reran the fast scripts. No fallback to another model occurred in any of these sessions. The manuscript has not been peer reviewed by humans.

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